

Personalized Product Recommendation Prediction

Abstract

Our project is **Customer Purchase Prediction and Personalized Product Recommendation using Machine Learning**. The main objective is to predict which products a customer is likely to purchase and recommend relevant products based on their shopping behavior. We use information such as purchase history, browsing activity, product categories, average spending, ratings, and purchase frequency. First, we preprocess the data by handling missing values, encoding categorical features, scaling numerical features where required, and selecting important features. We then implement and compare Logistic Regression, Decision Tree, and Random Forest algorithms for purchase prediction. We also use K-Means Clustering to group customers with similar shopping behavior. Finally, the system uses these predictions and customer segments to provide personalized product recommendations.

Research Paper – Main Points

Research Paper: E-commerce-based Smart Recommendation System using element-by-element collaborative filtering following with the Machine Learning Technology

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1. Main Idea

The paper presents a smart e-commerce product recommendation system using machine learning and collaborative filtering. The system uses user-related information, product data, purchase history and search behavior to provide personalized product recommendations.

2. Problem Addressed

E-commerce platforms need to provide relevant products to users from a large number of available products. The paper focuses on improving personalized recommendations and user experience.

3. Proposed Method

The proposed system uses an element-by-element collaborative filtering approach. Recommendations are generated by analyzing users' purchases and searches and identifying useful product relationships.

4. Key Technology

Collaborative Filtering – used to recommend products based on user behavior and preferences.

5. Outcome

The paper demonstrates a functional smart recommendation system intended to improve user engagement, shopping experience and product recommendation effectiveness.

6. Relevance to Your ML Project

This paper can be used as the research paper/reference for your Personalized Product Recommendation Prediction project. Both focus on analyzing customer behavior and recommending relevant products.

7. Research Gap / Your Project Difference

Your project extends the recommendation idea by combining purchase prediction, customer segmentation and personalized recommendations based on shopping behavior.